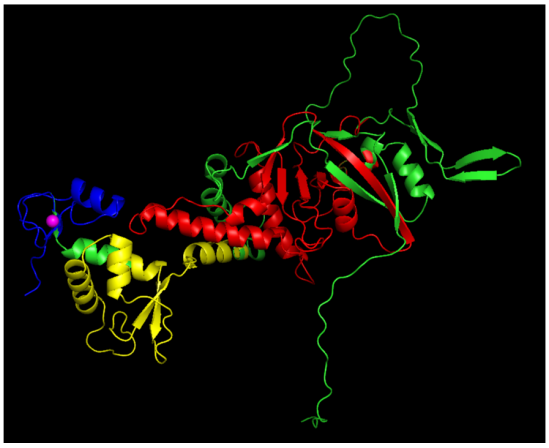
Quick Report: initial findings on aRES

E. coli UC5 aRES, 459 aa. Sequence composition and AlphaFold3 model analysis only. No experimental data. October 13, 2026



Zn ribbon 1–48 blue, NTD-1 49–120 yellow, RES 126–447 red.

Charge composition, Lys+Arg against Asp+Glu:

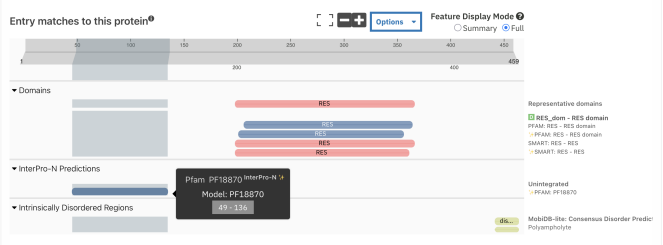
Region	Len	K	R	D	E	Net (/100)
Zn ribbon 1–48	48	10	3	2	0	**+11** (+22.9)
NTD-1 49–120	72	0	3	11	8	**−16** (−22.2)
N-block 1–120	120	10	6	13	8	−5 (−4.2)
RES side 126–447	322	23	19	24	21	−3 (−0.9)

The two N-terminal modules carry equal and opposite charge density. The ribbon has ten lysines and no glutamate. NTD-1 has nineteen acidic residues and no lysine. The RES domain is at −0.9 per 100, so the extremity is confined to the N-block.

Residues 31–40 are `CSYC KRKRK N`, a basic run directly after the second CxxC.

Two N-terminal domains

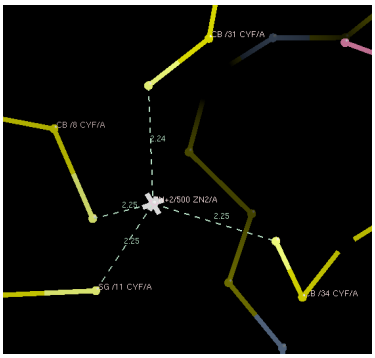
1. NTD-1



Domain	Range	Note
PF18870	49–136	no experimental structure exists
PF08808 RES	199–363	the NADase

2. Zinc ribbon

Cysteines	**C8, C11, C31, C34** (spacing 3, 20, 3)
Pairwise SG–SG (metal-free)	**3.55–3.81 Å**, mean **3.67 Å**
Zn–S (metal-bound)	**2.24–2.25 Å**
Ideal $Zn(Cys)_4$ edge	**3.76 Å**, deviation **0.09 Å**
Mean pLDDT of the 4 Cys	**90.8**
Not disulfides	an S–S bond is 2.05 Å
Control (T7 gp5)	10 Cys, **no cluster**; closest 7.85 Å



$Zn(Cys)_4$ coordination in the metal-bound model.

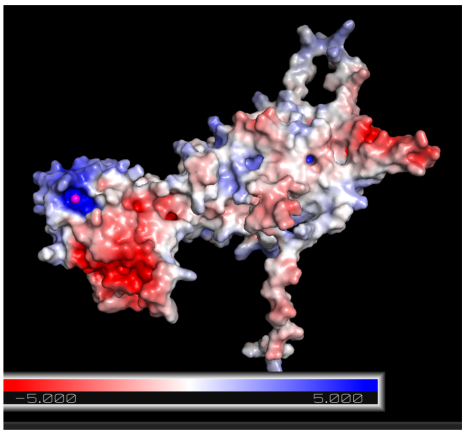
Together they form a bipolar surface

The basic ribbon face and the acidic NTD-1 face sit on adjacent lobes and each faces solvent. They do not neutralise one another.

Working model (hypothesis). The N-block recognises a polymerase engaged with nucleic acid:

- the basic zinc ribbon binds the DNA;
- the acidic NTD-1 binds the polymerase's basic cleft as a DNA mimic (cf. Ocr, T7 gene 0.3).

Consistent with recognition of unrelated polymerases, with λ (no phage polymerase), and with the DNA-free *in vitro* activation.



Electrostatic surface, $\pm 5\, k_B T/e$.

Caveats. Composition and one predicted model. The rendering may overstate the basic patch at the zinc if the four Cys were not treated as thiolates. One orientation. gp5 D106 is itself acidic. **Next:** residue-type surface and Zn-deleted control; is D104 on the acidic face; AF3 cofold with a primer–template duplex; EMSA 1–120 vs 49–136; DNA competition of the aRES–gp5 pull-down.